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Tech Skills

Foundation Models

Post-Training: RL (Sync/Async, On/Off-Policy) / MOPD, Reasoning Models, Custom Environments (RLVR, LLM-as-a-Judge), Large Scale Distributed Training with Multi-Dimensional Parallelism (x100+ GPUs), Inference Optimization

Infrastructure & Tools

vLLM, vERL, PrimeRL, HF Stack, SLURM, Ray, Proprietary Clusters, Docker, Kubernetes, CI/CD, Git, AWS/GCP

Classic Deep Reinforcement Learning

Online, Offline, Imitation / Curriculum / Adversarial Learning, League Training

Computer Vision

Object Detection, Semantic Segmentation, Behavior Classification

Modeling and Simulation

Multi-Body Dynamics, Atmospheric & Space Flight Dynamics, Robotics Systems, Sensor Fusion

Soft Skills

Strategic Thinking

Aligning development with real-world impact

First-Principles and Out-of-the-Box Thinking

Reasoning from fundamentals, approaching problems with unconventional perspectives

Creative Problem Solving

Tackling open-ended, high-uncertainty challenges

Cross-Functional Collaboration

Bridging engineering, research, and domain experts

Self-Directed Leadership

Driving projects from concept to deployment

Technical Communication

Explaining complex topics to diverse audiences

Alessandro Palmas

Senior ML Research Engineer

Senior ML Research Engineer with 15+ years of experience developing intelligent systems across foundation models, reinforcement learning, simulation, and multimodality. Currently focused on large-scale post-training of foundation models, designing and engineering distributed RL systems, verifiable training environments, reasoning benchmarks, and inference infrastructure across hundreds of GPUs. Experienced in turning emerging research ideas into scalable, production-grade training and evaluation systems, with deep expertise spanning RL post-training, reasoning models, distributed training, RLVR, custom environments, and inference optimization. Proven track record of bridging frontier AI research and large-scale engineering to develop capable, controllable, and reliable intelligent systems.

Experience

2025/10 - present

LawZero - Montreal • Permanent Full-time

Member of Technical Staff

Early employee and part of the core research and engineering team at LawZero, a non-profit AI research lab led by Yoshua Bengio that raised 300M+ USD to date. The lab's mission is to advance foundation models that are truthful, transparent, and safe-by-design, systems that understand the world rather than merely act in it.

- Responsible for RL-based post-training, orchestrating large-scale distributed training on proprietary clusters using multi-dimensional parallelism, optimized training/rollouts balancing, and async/off-policy mechanisms.
- Large-scale (hundreds of GPUs), distributed reinforcement learning for foundation models fine-tuning, leveraging verifiable environments to train models with controllable reasoning efforts across multiple domains covering natural language, math and code.
- Design and implementation of benchmarking frameworks to evaluate model reasoning and safety of different reference baselines for scalable oversight.
- Collaborating with cross-disciplinary teams to shape the lab's technical direction and infrastructure.

2022/12 - 2025/10

Ubisoft La Forge - Montreal • Permanent Full-time

Senior AI/ML Research Engineer

I led the development of next-generation ML systems for AAA games, with a focus on Deep Reinforcement Learning and simulation-driven agent training. My work spanned research, tooling, and deployment, delivering scalable learning pipelines and integrating AI behaviors into live game environments.

- Built and deployed Deep Reinforcement Learning systems for integration into AAA game environments.
- Designed scalable curriculum and multi-agent RL algorithms including League Training.
- Worked on Online, Imitation, Offline, and Hierarchical RL, explored Multimodal and Foundation Model Agents.
- Integrated AI models into proprietary engines for NPC behavior, testing, and live simulations.

Education

MSc by Research - Post-Grad Space Engineering
University Of Glasgow (2013)

MSc Space Engineering
Milan Polytechnic (2010)

MSc Aerospace Engineering
Turin Polytechnic (2010)

BSc Aerospace Engineering
Turin Polytechnic (2008)

Publications

[Linkedin](#) | [Google Scholar](#)

Language

Italian (Native)

English (Professional)

Spanish (Intermediate)

French (Intermediate)

Interests & Passions

Building AI-based Tech

Open Source

History of Mathematics

2018/02 - present

Artificial Twin • Freelance

AI/ML Consultant

Providing consulting and development services focusing on advanced AI/ML systems across industries, with a focus on Deep Reinforcement Learning, Foundation Models, computer vision, and physics-based modeling.

- Designed and delivered agents trained for decision-making and control through custom RL pipelines in simulation and real-world settings, leveraging curriculum learning, imitation learning, and hybrid modeling.
- Explored the integration of transformer-based LLMs and emerging Vision-Language-Action Models into end-user applications for reasoning, perception, and control.
- Delivered computer vision systems for detection, segmentation, and semantic understanding in industrial and robotics domains, optimized for deployment on constrained compute environments.
- Developed and maintained proprietary libraries supporting ML (RL, CV, LLM), physics-based simulation, and spatial intelligence workflows.

2020/11 - 2024/12

DIAMBRA (Acquired in Dec 2024) • Part-time

Founder & Principal AI/ML Engineer

Founded and led DIAMBRA, a cloud-native platform for competitive multi-agent RL. Enabled researchers and practitioners to train, benchmark, and deploy agents in complex environments.

- Built a distributed platform for tournament-style benchmarking with support for cloud inference, model submission, and automated evaluation.
- Developed cross-platform environments based on competitive fighting games. Supported workflows for curriculum design, human-in-the-loop training, imitation learning, and agent-vs-agent/human competition.
- Released tools as open-source Dockerized packages, fostering a community of RL researchers and ensuring reproducibility and real-world applicability.
- Led DIAMBRA to acquisition, validating the product's technical foundation and commercial relevance.

2017/07 - 2022/11

Nurjana Technologies • Permanent Full-time

Principal Research Engineer

Led development of AI-driven autonomous systems, perception, and simulation for aerospace and defense applications. Delivered real-world solutions across aerial ISR, satellite collision avoidance, and robotics, integrating ML, embedded systems, and physical modeling.

- Deployed real-time computer vision models (object detection, segmentation, tracking, activity recognition) on edge devices (e.g., NVIDIA Jetson) for drones operating in GNSS-denied zones.
- Built hardware-in-the-loop pipelines with Unreal Engine to validate vision-based navigation and target detection in photorealistic environments.
- Developed custom flight control, vision-based landing, and obstacle avoidance for autonomous aerial platforms. Integrated vision, control, and simulation into deployable systems.
- Designed software for orbit propagation, attitude dynamics, conjunction analysis, maneuver planning, and sensor tasking. Contributed to operational space situational awareness systems.